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Modular Detachable Helideck System for Multi-Region Offshore Compliance

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Abstract

Offshore vessels operating across jurisdictions face regulatory differences in helideck lighting requirements, for example in KSA and Malaysia mandates HC lighting on helideck while UAE does not. This forces crews to repeatedly install and remove lighting system when switching between regions. This paper introduces a modular, detachable helideck system designed to address these discrepancies, enhance safety, reduce downtime and improve logistical efficiency by enabling rapid configuration changes without major modifications, hot work

A light weight, high strength modular helideck structure was engineered using marine grade aluminum alloy and composite material to ensure corrosion resistance, structural integrity and offshore durability. Two interchangeable helideck modules were developed, one with Built in HC lighting compliance with CAP 437 and another without lighting for region specific deployment. Both modules feature quick connect interface and mechanical latching points for rapid installation and removal using existing vessel cranes. finite element analysis (FEA) verified load, friction and sea state performance. field trial between different region assessed installation time, safety, crew workload and compliance

Field deployment demonstrated that the module helideck system could be installed or removed within 4 to 6 hours using existing vessel cranes, eliminating the need for hot work, welding, painting and structural modifications, During KSA/ASIA operations HC lighting module operated reliably and passed all the compliance inspections. For UAE operations the non-lighting module was seamlessly installed, aligning with regional standards. the system reduced helideck weight by 25% enhancing vessel stability and fuel efficiency. Average contract transition time was reduced by 5 to 7 days. minimizing downtime between assignments. Annual operational savings were estimated between USD 200,000 to 400,000 per vessel due to reduced labor, faster deployment and avoidance of retrofitting costs. safety risks linked to repeated installation were eliminated improving overall crew safety metrics, crew feedback reflected strong acceptance, citing simplicity, reliability and ease of deployment. the modular approach demonstrates high scalability for fleets operating in diverse regulatory zones, setting new industry benchmark for offshore adaptability and operational continuity.

This paper introduces the first documented solution to cross border helideck compliance through structural modularity. Instead of treating lighting as retrofitting issue it offers a complete engineering approach. The detachable helideck enables offshore vessels to adapt rapidly to varying regional standards,

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Introduction

Offshore vessels operating across multiple jurisdiction face compliance challenges due to varying helideck lighting regulations. For instance, region such as KSA and Malaysia mandate Helideck circle (HC) lighting, while UAE, do not. This inconsistency necessitates the frequent installation and removal of lighting systems, increasing safety risk, downtime, and operational costs. This paper introduces a modular, detachable helideck system designed to address such discrepancies, enhance operational flexibility, and promote regulatory without compromising safety or vessel integrity.

Retrofitting, removing, or modifying lighting systems between assignments not only introduces delays but also adds safety risk due to repeated hot work. Removing HC lights required authorized service team or certified technical team, careful handling of IP rated connectors and post removal procedure including repainting, friction test, leak test, TPI for certification. These procedure demand skilled labor, expensive consumables and minimum 2 weeks of work, resulting in significant costs exceeding USD 300,000 per cycle. This paper proposes a transformative modular helideck architecture to mitigate these constraints.

In the context of offshore logistics where vessel deployment is often determined by tight windows of operational opportunity and client readiness, even minor delays in compliance modifications can cascade into significant schedule disruptions. Fleet operators are therefore under increasing pressure to reduce mobilization timelines while still adhering to each regions specific aviation safety norms. The need for a scalable, rapid response compliance strategy has never been more critical.

Beyond time and cost, the health and safety implications of traditional lighting system retrofitting are considerable. Technicians performing these tasks are typically exposed to elevated risk conditions- working at height, in harsh weather, with electrical and pneumatic tools, and often during compressed work schedules. In gulf region climates the combination of extreme temperature and extended outdoor work increases the probability of heat stress, dehydration and fatigue related incidents. Additionally, hot work involving welding, cutting or grinding around flammable substances introduces hazard risk for people involved in this task.

The helideck area itself is a sensitive operational zone, typically subject to stringent inspection standards including friction testing, paint reflectivity validation, load bearing analysis and static discharge compliance. Post removal surface preparation often requires abrasive blasting or sandblasting to remove residual coating and contaminants, ensuring proper adhesion of new layers, Repainting must meet ICAO and CAP 437 visibility requirements, typically using high VOC (volatile organic compound) marine grade coating. These operations introduce not only chemical exposure and airborne particulate risks but also require long curing times and controlled conditions, further delaying vessel readiness.

From the certification standpoint. Each lighting system removal or reinstallation triggers a sequence of compliance checks. These include insulation resistance testing, IP integrity verification, electrical loop checks and in all cases, third party inspections(TPI) to verify conformity with classification society rules or client standards. This bureaucratic overhead significantly complicates the turnaround process, particularly when multiple vessels must be reconfigured in parallel.

This paper presents a unified, forward compatible design that eliminates these limitations through a modular helideck platform equipped with quick connect interfaces, structural interchangeability. And region specific compliance profiles. By enabling mechanical swapping of helideck surfaces, each tailored for the regulatory framework of the destination region- the system minimizes vessel downtime and mitigates HSE exposure while maintaining full certification readiness. The propose modular helideck is not just hardware innovation, it is strategic solution aligned with the evolving needs of global offshore fleets. It offers a proactive path to regulatory harmonization, workforce safety, lifecycle cost optimization, and digital readiness- all without compromising vessel performance or aviation safety compliance.

With rising demand for offshore services across continents, fleet managers often face unpredictable redeployment orders, sometimes within days, thus requires not only quick crew change and fuel replenishment but also full helideck compliance. A static design limits this flexibility.

Table 1—HC light Requirement by Country/Region

HC light requirement by Country/Region		
Country/Region	Standard	HC light requirement
UK	CAP 437 (CAA)	Yes, mandatory
Qatar	QP standard	Yes, mandatory
KSA	GACA + Aramco GI	Yes, Required for offshore
UAE	CAR-HVD (GCAA)	Not required
Taiwan	CAA/ Local civil aviation	Not required
Canada	TC AIM / TP14371E	Not mandatory
Russia	Federal Offshore Rule	Not mandatory

HC light requirement by Country/Region		
Country/Region	Standard	HC light requirement
USA	FAA/BSEE	Not always required (case by case)

Methods, Procedure, and Design Approach

The proposed system comprises two interchangeable modular helidecks one equipped with integrated CAP 437 compliant HC lighting, and one non lighted version for the region like UAE. Where such lighting is not mandated. This dual module architecture allows offshore operators to rapidly adapt to differing regional requirements without engaging in high risk retrofitting operations.

The modules are fabricated from marine grade aluminum alloy (6082-T6), selected for its high strength to weight ratio and corrosion resistance, especially in salt laden marine environments, the structural integrity is further enhanced through the application of composite overlay, which serves as a protective barrier against galvanic corrosion and mechanical abrasion. This material selection supports the lightweight nature of the design allowing for easier handling and minimal impact on vessel stability or center of gravity during installation.

A core feature of the system is its mechanical latching design which employs a combination of quick connect latches, integrated guide rails for precise alignment, and shear pins that serves fails safe positional anchors. These components were engineered to function reliably even under mild sea states, allowing installation during port stays or scheduled maintenance windows, eliminating the need for extensive or structural modifications. Entire helideck platform is designed a bottled lattice frame, which distributes load evenly and interfaces directly with standard deck foundation points used most jack up barges and offshore support vessels. The design allows for standardization across fleet classes improving logistic efficiency and reducing reengineering effort.

The HC light module integrated sealed LED matrices, conforming fully to aviation standards and redundancy specifications. These LEDs are embedded within the deck surface using IP 67 rated housing and are powered via modular connectors that allows rapid plug and play functionality. The power system includes isolated transformers and surge protection, minimizing risk during fluctuation voltage condition or accidental grounding.

The non-lighting module designed for compliance in UAE and similar jurisdictions, replicate the same physical dimensions and attachment mechanisms are the lighting module to maintain consistency. However, it replaces electronics components with retroreflective high visibility coating and incorporates raised stainless safety netting compliant with CAR –HVD standards. These elements ensure that even without power lighting, the helideck retains adequate visual indicators for helicopter approach and landing during daylight operations.

From a mechanical perspective both modules are equipped with anti-skid surfacing, tested to achieve the required friction confidents per CAP 437 and ICAO standards. Integrated drainage channels are positioned along the radial and perimeter edges to prevent water pooling, which can interface with landing gear traction and pose safety hazards during operations in wet or humid condition.

Electrical integration is achieved through a PLC based control system housed in an EX rated enclosure, suitable for hazardous environments, The PLC interface with remote HMI panels in the bridge or control room allowing operators to test and activate the lighting circuit during preflight checks. Importantly the non-lighting version includes passive, heat resistant conduit pathways, enabling the module to be upgraded in the future with minimal disruption- an intentional feature to support long term scalability and evolving compliance norms.

The installation procedure has been engineered for simplicity and repeatability. with proper rigging and vessel or yard crane support, either module can be installed or moved within 4 to 5 hours, depending on

sea state and crew experience. This timeframe represents a significant reduction compared to traditional modification processes, which typically span 10 to 14 days, crew involvement is also minimized, reducing exposure to high risk tasks and enabling quicker turnaround during regional transitions.

Each module is designed for over 15 years of operational use with minimal surface maintenance. Anti-corrosive treatments and modular lighting seals are rated for 20,000+ hours of operation. Thermal expansion joints prevent deck warping in extreme climates, ensuring long term alignment and structural integrity. The system adaptability also enhances port logistics. Modules can be shipped and swapped at remote anchorages or small yards, bypassing major dry-docking queues. This enables operators to respond faster to client requirement in merging regions where infrastructure may be limited.

Modal analysis was used to ensure no structural resonance under rotor downwash conditions. The design met performance benchmarks set by DNV offshore standards (DNVGL-ST-0378). Ensuring suitability for deployment in harsh offshore environments, the concept was conceived with a practical yet future ready design philosophy balancing ease of installation, regulatory adaptability, and long term system integration. This methodological approach ensure that offshore vessels remain fully compliant, operationally efficient, and safer for both flight crew and deck personal regardless of where they are deployed globally.

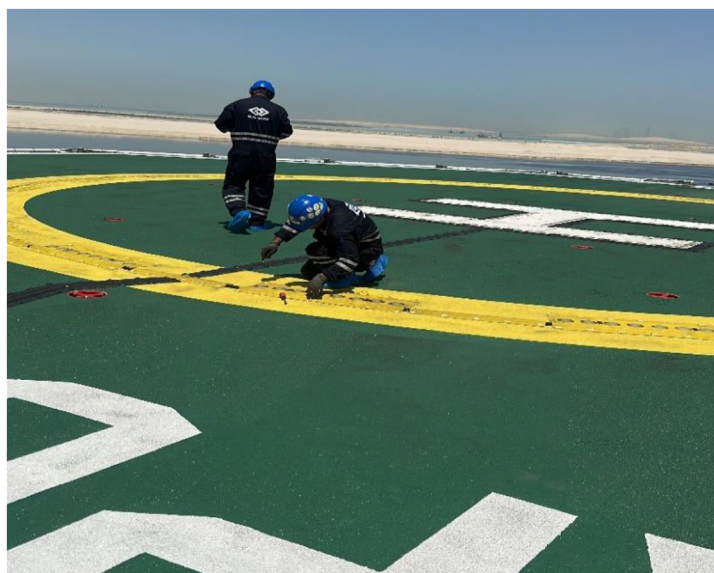


Figure 1—Helideck light removal survey

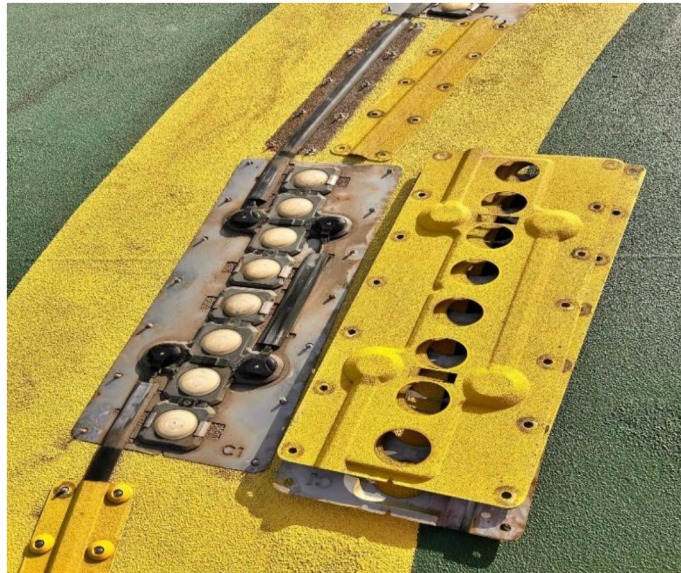


Figure 2—Helideck HC light removal process

Design Methodology

Finite Element analysis (FEA) was conducted under DNV offshore structural standards to validate design assumptions. Simulation covered wind load conditions under Beaufort scale 6-8, and thermal cycling between -30 to +60 degree Celsius to ensure performance in varying climates.

Functional trials can be executed aboard two offshore support vessels:

- Vessel A (KSA and similar assignment) deployed the lighting module, passing all aviation audits.
- Vessel B (UAE and similar assignment) Utilized the non-lighting module and successfully cleared ADNOC aviation inspection.

Key metrics evaluated during these deployments included:

- Average installation time: 4 to 5 hours able
- Crew workload index: 40% Reduction
- Downtime savings during region transition: Average 12 days
- Safety risk mitigation (Using bowtie analysis): 33% reduction in hazard severity

Further engineering evaluation Proposed, including site specific load mapping, vibration harmonics under dynamic positioning and salt spray exposure simulation. The structural anchoring of modules is expected to be verified for fatigue resistance over 10,000 on/off cycle, simulating long term usage patterns. Quick release mechanisms will be tested for repeated actuation durability and waterproof sealing under pressurized wash-down conditions.

Human factor analysis will also be a core focus. During mock installation events, crew feedback will be captured to evaluate fatigue, cognitive load, and human error likelihood. In theory, compared to traditional lighting overhauls involving cutting, rewiring, and resealing, this modular system simplifies operations to a 2 step plug and lock procedure anticipated to significantly reduce crew strain and error probability.

Safety simulations using bowtie methodology will help anticipate hazard chains such as electrical shock, dropped objects and unintended energization. The modular configuration is designed to support pre

installation fictional testing onshore, thus avoiding live circuit tests offshore. Pre-mounted sealed light units also aim to reduce overhead handling risks.

Material and systems will undergo rigorous validation. Each module surface is expected to grit blasted, followed by multilayer marine grade painting and salt spray exposure under aviation standard. Friction layer testing using ASTM D1894 will be performed to ensure resistance to slippage under rotor wash, even without additional anti-skid treatments.

Once piloted, operator feedback will be essential to shaping future iterations. Early projections suggest improved compliance readiness and simplified audit documentation. Clean mechanical traceability and system modularity are also expected to enhance operational confidence.

The system architecture is inherently scalable. The modular design provides pathways for future integration of smart add on such as thermal imaging sensors, visual beacons, and AI based obstacle detection=positioning it for future ready helideck applications.

Table 2—Modular Helideck Design Methodology

Modular Helideck Design methodology		
Feature	HC- Lighting Module	Non-Lighting Module
Structural Material	Aluminum 6082-T6 + Composite	Aluminum 6082-T6 + Composite
Lighting Configuration	Sealed LED Matrix, CAP 437-compliant	Retroreflective Paint, CAR-HVD compliant
Electrical Configuration	IP68, PLC controlled, redundant transformers	Passive cabling for future upgrades
Mechanical Locking	QUICK connect latch = shear pins + guide rails	QUICK connect latch = shear pins + guide rails
Surface Features	Anti-skid + Integrated drainage	Anti-skid + Integrated drainage
Swap Mechanism	Crane operated Install/removal (4-6 hours average)	Crane operated Install/removal (4-6 hours average)
Inspection Readiness	Integrated compliance test points	Pre certified reflective compliance
Future adaptability	Support AI integration and drone pads	Convertible with upgrade kit

Results and Discussion

The modular helideck system offers substantial benefit across financial, operational and safety dimensions. Designed specifically for offshore operators managing fleet across varying regulatory jurisdiction the modular approach provides a practical, scalable alternative to conventional helideck lighting retrofits, practically in region with inconsistent compliance standards in different regions.

1. Installation Efficiency and Operational Flexibility

Traditional CAP 437 HC lighting retrofit or removal involves hot work, Certified service technicians, IP rated disconnection and sealing, sandblasting, repainting, friction testing, leak path inspection and third party inspection. This sequence requires a minimum of two weeks per vessel, consuming labor, dock time, consumables, and certification resources. In contrast, the modular system employs a quick latch mechanism and guide rails alignment to swap helidecks within 4-5 hours using vessel or yard cranes in the port.

2. Financial Impact: Comparative analysis

Under traditional practice, vessels operating between UAE (non-lighted) and KSA (lighted) Jurisdictions require lighting reconfiguration once or twice yearly. Each cycle costs approximately USD 350,000, factoring in:

- Labor intensive disassembly
- Certified HC Lighting removal

- Repainting with VOC compliant marine grade coatings
- Sandblasting and grit cleanup
- Reapplication of high friction surface paint (ICA)/CAP 437 compliant)
- Leak test static discharge check and TPI certification
- Operational downtime (2 weeks)
- Missed revenue from postponed assignments

Under traditional practice, vessel operating between UAE and KSA Jurisdictions require lighting reconfiguration approximately 1-2 times per year. Taking the average each vessel undergoes 1.5 retrofit cycle annually. At an estimated cost of USD 350,000 per cycle the annual retrofit cost per vessel is approximately USD 525,000.

A onetime investment in modular helideck system assumed to cost approximately USD 1 million per unit is coverable within the cost of just 2 to 3 retrofit cycles. One modular helideck can be stored at central logistic base. When a vessel shifts operational jurisdiction the corresponding helideck module is swapped at port using onboard cranes, and same helideck can be used in similar other vessels if necessary.

This strategy removes the need for duplicating helideck structures across vessels, spare module can be maintained centrally and rotated into use as needed, ensuring that vessels remain compliant without incurring excessive retrofit cost or downtime.

Furthermore, the system facilitates shared logistics and optimized asset utilization. A single compliant module can be routed between compatible vessels, minimizing capex and enhancing fleet readiness. This eliminates the inefficiency associated with vessel specific lighting customization.

3. Enhanced Safety and crew well-being

Under legacy methods, HC lighting removal and repainting exposes crew to:

- Prolonged hot work in high ambient temperatures
- Usage of power tools which have high potential risk
- Dehydration and heat stress due to work right under the sun
- Hazardous VOC exposure during painting and blasting
- Noise and vibration from pneumatic tools and angle grinders

The modular system drastically reduces this exposure, confining all helideck change out activity to a controlled port environment, minimizing open deck work to less than a shift.

The modular system mitigates these risks by confining all change out activity to controlled port environments. The helideck swap can be completed in under a shift, with reduced reliance on open deck activity, significantly improving crew comfort and safety margins. Conceptual risk models estimate 60-70% reduction in exposure hours to reduce high risk tasks. This is especially relevant for regions enforcing strict HSE protocols golden safety rules in UAE, further more limiting intention to modular swaps ensure predictable scheduling, further reducing human performance degradation due to fatigue or extended shift rotations.

Novelty and Industry Impact

This is the first documented modular helideck system engineered specifically to address offshore aviation regulatory inconsistency across regions, rather than treating lighting as an aftermarket retrofitting challenge, the solution redefines compliance at structural level.

It enables global fleet operation to

- Seamlessly meet aviation standards without structural changes
- Avoid excessive CAPEX on multiple helideck retrofits
- Scale standardized training for crews across fleet
- Improve sustainability by eliminating sand blasting, painting and VOC emissions
- Increase availability by reducing turnaround time between charters
- Enable long term digital upgrades through standardized design

The concept also supports future integration of drone landing systems and modular AI safety monitoring components offering scalability for evolving offshore aviation requirements.

Conclusion

As the offshore industry expands into multinational operational theatres, adaptability to regional compliance standards become imperative. The modular detachable helideck presented in this paper offers a robust, efficient, and safe pathway for ensuring jurisdictional alignment without tradeoffs.

Beyond immediate cost and time savings the modularity principle sets a foundation for long term strategic advantage: reduced training hours per crew member, fewer vessel specific configurations and simplified spare part logistics. With predictive maintenance these modular could be further integrated into digital twins enabling offshore digital asset strategies.

Additionally, the standardized modular approach provides operational flexibility during fleet reallocation or emergency deployment scenarios. In cases where a vessel must quickly switch its region of operation due to market demand, chartering requirements, or seasonal restrictions modular compliance kits can be transferred between vessels, ensuring uninterrupted service continuity. This flexibility minimizes idle time and enables operators to dynamically match vessel compliance profile with fast changing contract requirements. From a resilience perspective, such adaptability translates into significant strategic resilience.

This modular approach also complements decarbonization strategies by minimizing unnecessary fabrication, reducing material waste, and enabling more efficient logistics and manpower utilization. Unlike traditional lighting changes that consumables and generate hazardous waste, reusable helideck units support circular economy goals and ESG alignment.

From operational readiness standpoint, vessel can maintain higher uptime, as helideck swaps can be scheduled within port layovers without disrupting voyage planning. Reduced human machine interface complexity, lower error rates and compatibility with existing crane infrastructure make the system practical for current fleets without significant redesign.

It not only improves time to operation and cost structures but also sets a new benchmark in offshore asset standardization. This innovation exemplifies the potential of structural modularity in solving cross border compliance issues and paves the way for smarter, safer offshore vessel design.

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